

(1) Primordial Multi-Planet Architecture

Coplanar circular orbits ($e \approx 0$)



Hill unstable separation $\Delta < 3.46 R_{H, \text{mut}}$

(2) Resonant Overlap & Orbit Crossing

Chaotic resonance overlap ($P_1/P_2 \sim p/q$)



Close encounter distance $d \lesssim R_{H, \text{mut}}$

(3) Strong Gravitational Scattering & Ejection

Ejected planet m_2 ($E > 0$, $v_\infty > 0$)



Surviving planet m_1 : $a_f = a_1 / (1 + m_2/m_1)$

(4) Relaxed Eccentric Orbit & Hot Jupiter Migration

Surviving eccentric planet ($E < 0$, $v_\infty = 0$)



Tidal Circularization if $q \leq 0.05 \text{ AU} \rightarrow \text{Hot Jupiter } (a_{\text{circ}} \approx 2q)$